

Total No. of Questions : 8]

SEAT No. :

PC-2498

[Total No. of Pages : 3

[6354]-627

B.E. (Mechanical)

QUALITY AND RELIABILITY ENGINEERING
(2019 Pattern) (Semester - VIII) (Elective - V) (402050A)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Attempt Q1 or Q2, Q3 or Q4, Q5 or Q6, Q7 or Q8.
- 2) Draw suitable neat diagrams, whenever necessary.
- 3) Figure to the right indicate full marks.
- 4) Assume suitable data if required.

Q1) a) Define the following (Any 4):

[8]

- i) Failure Rate
- ii) Hazard Rate
- iii) Mean Time to Failure (MTTF)
- iv) Failure Density
- v) Availability

b) Explain importance of reliability and what are the quality & reliability assurance rule?

[9]

Q2) a) Explain uncertainty and its types.

[8]

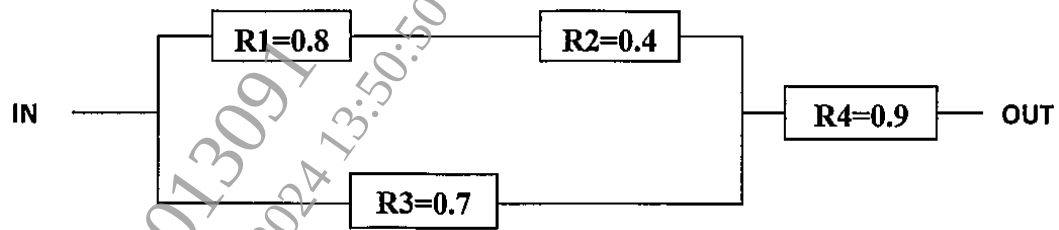
b) Following table shows test results of 1000 components, tested simultaneously. Evaluate: Hazard Rate, Failure density function and reliability.

[9]

Operating time	0	100	200	300	400	500	600	700	800	900	1000
Number of surviving components	1000	895	810	730	660	600	545	495	450	410	373

P.T.O.

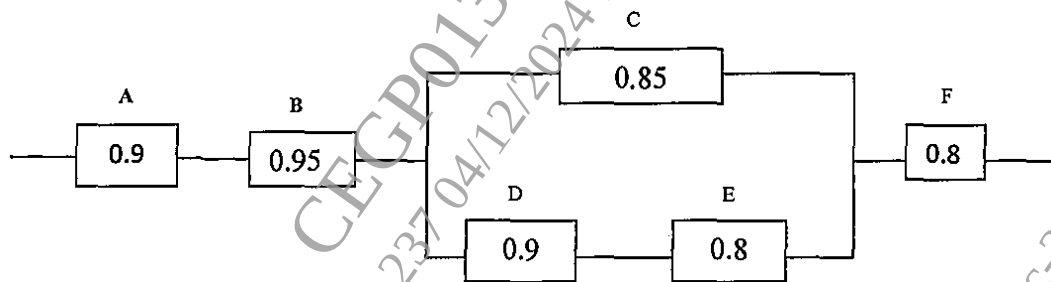
- Q3) a)** A network made of four independent units representing system is shown in figure. The reliability of R_i of unit- i for $i=1, 2, 3, 4$ are Given. Calculate the network reliability by using mixed network reduction method. [10]



- b) Consider system composed of three sub subsystem with estimated failure rate of $\lambda_1 = 0.05$, $\lambda_2 = 0.003$ and $\lambda_3 = 0.001$ failure per hour respectively. The system mission time of 20 hours. A system reliability of 0.95 is required. Find the reliability requirement for the subsystems. [8]

OR

- Q4) a)** A system consists of six components connected as shown figure. [10]

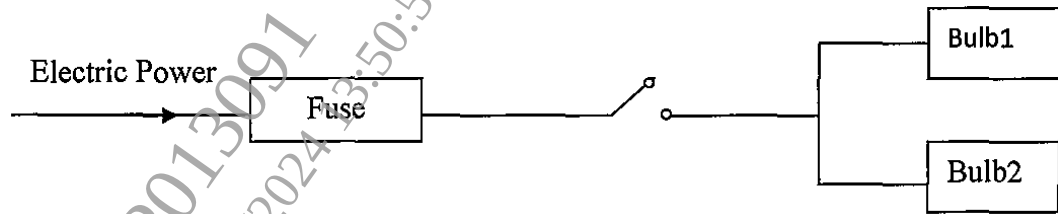


If the system reliability is to be improve value of 0.85 for period of 10 hours, Determine the reliability goal of each component by using minimum effort method.

- b) Explain in details any two system reliability models. [8]
- Q5) a)** Explain tie set and cut set method of reliability evaluation. [8]
- b) What is purpose of constructing the fault tree diagram? Explain the symbol used while constructing the fault tree diagram. [9]

OR

- Q6) a)** What is FMEA? Explain procedural steps involved in FMEA. [9]
- b)** A room with two light bulbs is operated by a single switch. By assuming the no light in the room as the top undesirable event construct the fault tree diagram. [8]



- Q7) a)** Write short notes on (Any two) : [8]
- Maintainability
 - Reliability Centered Maintenance (RCM)
 - Objectives of maintenance
- b)** A Beam is subjected to mean stress 180 N/mm^2 and standard deviation 20 N/mm^2 . The mean strength of material is 280 N/mm^2 and standard deviation 40 N/mm^2 . [10]

(Given: for $Z = -2.24$, Area = 0.4875)

Determine :

- Reliability of Beam
- Minimum factor of safety
- Average factor of safety.

OR

- Q8) a)** Difference between Highly Accelerated Life Testing (HALT) and Accelerated Life Testing (ALT). [8]
- b)** In short sample accelerated life testing of system based on weibull distribution the following data are recorded. [10]

Failure no.	1	2	3	4	5	6	7
MTTF (hrs.)	28.0	12.0	21.5	26.0	35.0	38.0	30.0

Plot the variation of reliability against time using:

- Mean Ranking
- Median ranking method

